FoilsMode

5D Bayesian Optimization of the Mu2e Stopping-Target Foil Stack

Y. Oksuzian

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Mu2e — autoresearch / closed-loop BO

Motivation

- Does modifying the **stopping-target foil stack** buy CE efficiency?
- Open a BO line over **extra foils added upstream / downstream** of the current deployed stack.
- Keep the deployed 37-foil base **pinned** — any movement in (sob, calo) is attributable to the +12 envelope.

The physical knob

Base 37-foil stack (deployed v02, pinned):

- `rOut = 75 mm`, `halfThickness = 0.0528 mm` (≈ 105.6 μm full)
- `holeRadius = 21.5 mm`, 37 foils, fixed spacing

The +12 envelope (optimized):

- ≤ 6 extra foils **upstream**
- ≤ 6 extra foils **downstream**
- All extras share **one** (rOut, halfThickness, rIn) triple

5D search space

Knob	Type	Range
`n_up`	Integer	0 – 6
`n_down`	Integer	0 – 6
`extra_rOut`	Real	50 – 250 mm
`extra_halfThickness`	Real	0.05 – 1.0 mm
`extra_rIn`	Real	0 – 50 mm

Objective: `obj = S/√B − α · calo / POT` ($\alpha = 1 \times 10^5$)

Pipeline

```
propose → preflight (G4 surface check)
       → mubeam → run1b_mubeam → concat → mustops_ce
       → harvest → scan_logs → leaderboard
```

- LangGraph state machine, q parallel children per round.
- GP refit between rounds (skopt EI, `cl_min` parallel strategy).
- Stop criteria: `--max-rounds`, Pareto-hash convergence, or operator `STOP_CLOSED_LOOP`.

Phase 0: preflight passes the extreme corners

Three extreme-corner configs hand-built and run through G4 surface-check **before any grid submit**:

name	n_up	n_down	rOut	hT	rIn	radii len	Result
`foilsP0_AU`	6	0	250	1.0	50	43	**PASS**
`foilsP0_AD`	0	6	250	1.0	50	43	**PASS**
`foilsP0_AS`	6	6	250	1.0	50	49	**PASS**

All three: `total_hits = 1, baseline = 1, managed = 0` — no overlap.

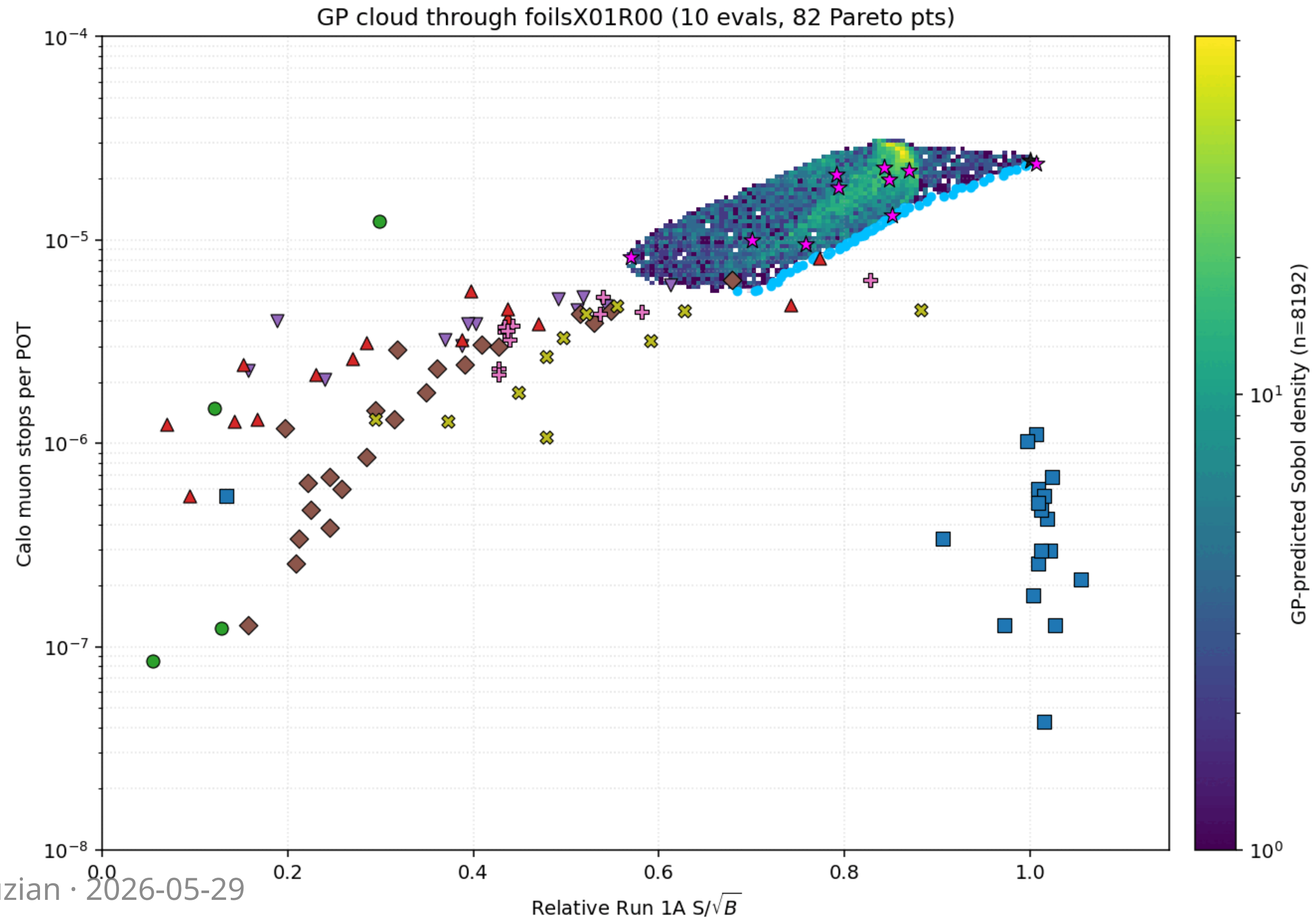
Full 5D box is buildable — no defensive clamp needed.

Run timeline

Run	q	Rounds	Evals	Status
X01	10	1	10	Sobol bootstrap, all PASS preflight
X02	10	3	30	clean, frontier still moving
X03	10	5	50	frontier widening, non-converged
X04	10	10	100	**in progress**

End of X04: ~190 evals total budget.

GP cloud evolution (animated)



Top configurations so far

config	n_up	n_down	rOut	hT	rIn	sob	calo ($\times 10^{-5}$)	obj
`foilsX03R04_02`	5	6	184.4	0.121	0.0	3.43	1.29	**2.14**
`foilsX03R03_03`	5	6	203.4	0.080	0.0	3.51	1.42	2.09
`foilsX03R04_00`	5	6	174.1	0.125	0.0	3.47	1.40	2.07
`foilsX03R02_06`	5	6	225.4	0.080	0.0	3.41	1.45	1.96
`foilsX03R02_03`	6	6	198.5	0.050	0.0	3.63	1.73	1.90

Pattern (top-5): max-extras dominates — `n_up ∈ {5,6}, n_down = 6, rIn = 0`.
Optimum sits in the **thin-foils / large-radius** regime.

Open questions / next steps

- **foilsX04** running now (100 evals, q = 10 × 10 rounds).
- If `extra_rIn` remains under-trained after X04: **hand-seed** small-rIn / large-rIn probes.
- Consider promoting a **6th dimension** (e.g. base hole radius decoupled from extras) if 5D plateaus.